Options & Commodity Derivatives Analytics Platform

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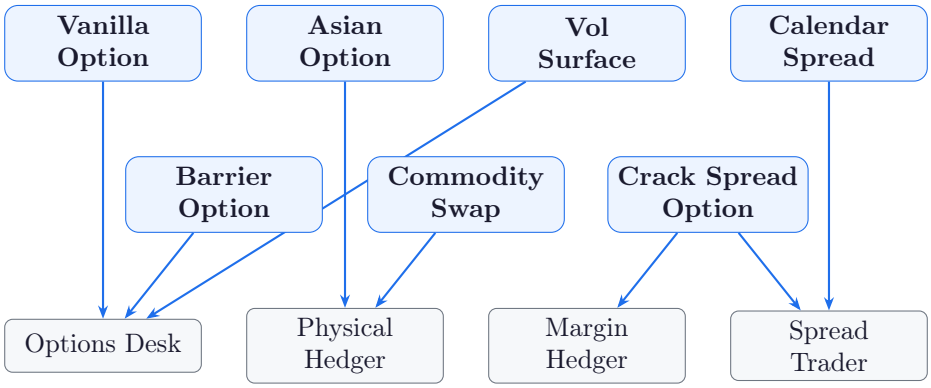
1 Introduction

1.1 Commodity Derivatives at a Glance

A commodity derivative is a financial contract whose value depends on the price of an underlying futures contract. The figure below illustrates the spectrum of products covered by OCDAP and their typical use cases:



1.2 What is OCDAP?

OCDAP (Options & Commodity Derivatives Analytics Platform) is a Python application that implements six industrial pricing engines connected to a live forward curve downloaded from Yahoo Finance or TradingView. It covers **65+ commodities** across 8 asset classes.

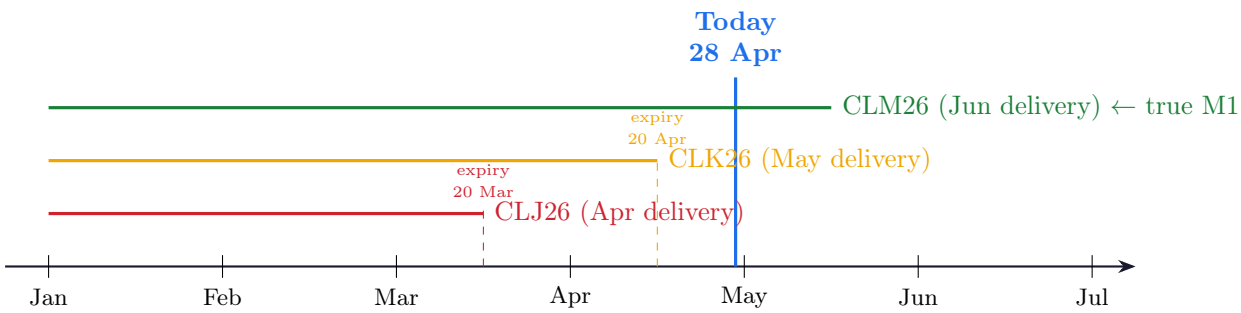
Tab	Product	Model
Vanilla Options	European call/put on futures	Black-76
Asian Options	Arithmetic/geometric average-price option	Monte Carlo + Kemna-Vorst
Crack Spread	Option on refining margin $F_2 - F_1$	Kirk (1995)
Calendar Spread	Option on curve slope $F_{\text{near}} - F_{\text{far}}$	Kirk on term structure
Commodity Swaps	Fixed-for-floating commodity swap	Discounted cashflows
Barrier Options	Knock-in / knock-out	Daily Monte Carlo
Vol Surface	Parametric implied volatility surface	Black-76 + skew
Forward Curve	Live futures curve + cost-of-carry extension	Yahoo Finance / TradingView

2 Architecture

2.1 The Pricing Pipeline

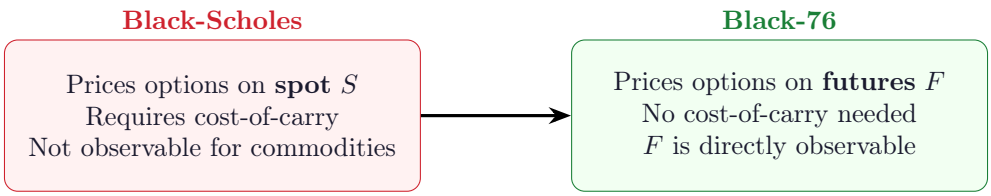
2.2 Expiry-Aware Ticker Generation

A critical design choice: OCDAP always downloads the **true front month**, not an expired contract. The timeline below shows why:



3 Black-76 Vanilla Option Pricer

3.1 Why Black-76 and Not Black-Scholes?

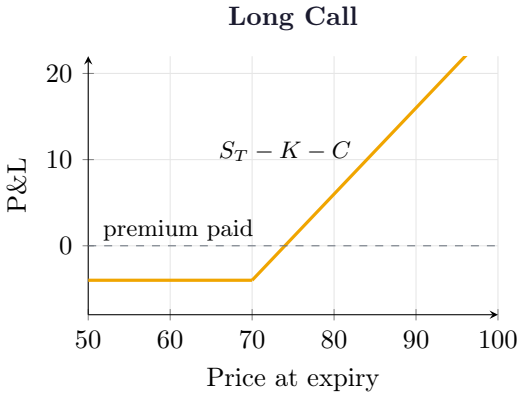


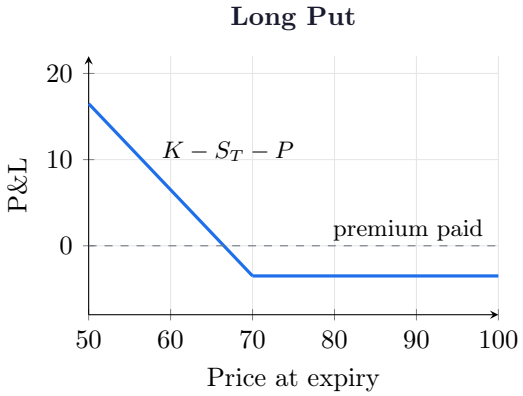
3.2 Pricing Formula

$$\boxed{C = e^{-rT}\left[F \cdot N(d_1) - K \cdot N(d_2)\right]} \qquad \boxed{P = e^{-rT}\left[K \cdot N(-d_2) - F \cdot N(-d_1)\right]} \tag{1}$$

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

3.3 Payoff Diagrams





3.4 Greeks

Greek	Formula (call)	Economic interpretation
Δ	$e^{-rT}N(d_1)$	Futures equivalent: how many futures to hold to delta-hedge
Γ	$\dfrac{e^{-rT}N'(d_1)}{F\sigma\sqrt{T}}$	Rate of change of delta — cost of dynamic rebalancing
$\mathcal{V}$	$\dfrac{e^{-rT}FN'(d_1)\sqrt{T}}{100}$	P&L per 1% implied vol move
Θ	$\dfrac{-e^{-rT}FN'(d_1)\sigma}{2\sqrt{T}\cdot 365}$	Daily time decay (negative for long options)
ρ	$\dfrac{-Te^{-rT}(FN(d_1) - KN(d_2))}{100}$	Sensitivity to risk-free rate per 1%

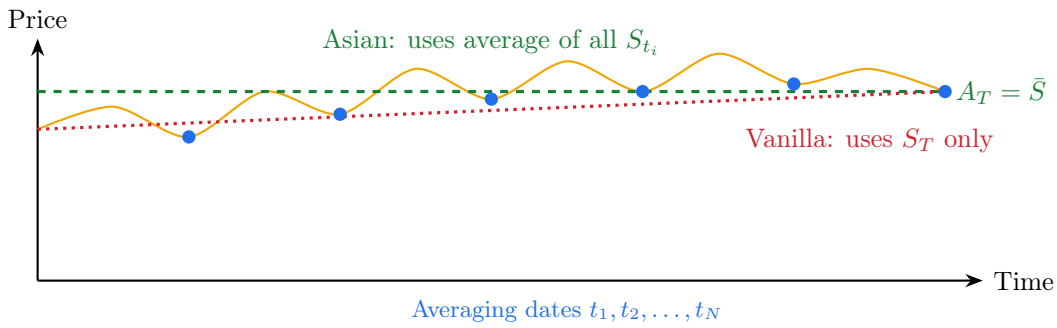
3.5 Put-Call Parity

$$C - P = e^{-rT}(F - K) \tag{2}$$

OCDAP verifies this numerically for every pair. The typical numerical error is below 10^{-4}, displayed as ✓ `verified` in the dashboard.

4 Asian Option Pricer

4.1 Why Asian Options in Commodity Markets?



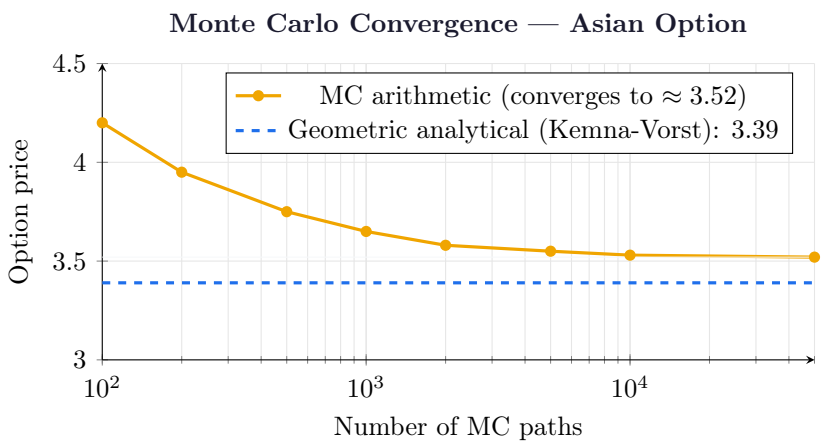
4.2 Monte Carlo Simulation

Under the risk-neutral measure, futures prices follow GBM with zero drift:

$$S_{t+\Delta t} = S_t \cdot \exp\left(-\tfrac{1}{2}\sigma^2 \Delta t + \sigma\sqrt{\Delta t}\, Z\right), \quad Z \sim \mathcal{N}(0,1) \tag{3}$$

$$C_{\text{Asian}} = e^{-rT} \cdot \frac{1}{M} \sum_{j=1}^{M} \max\left(\frac{1}{N}\sum_{i=1}^{N} S_{t_i}^{(j)} - K,\ 0\right) \tag{4}$$

4.3 Arithmetic vs. Geometric Average



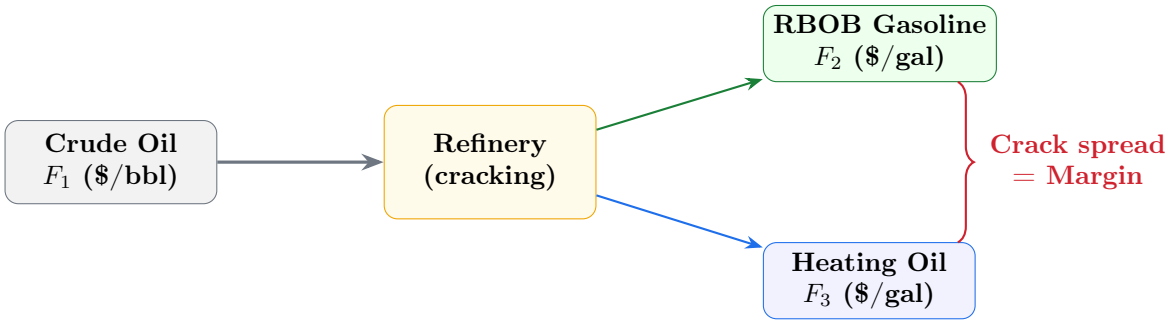
Ordering property (AM-GM inequality): The arithmetic average always exceeds or equals the geometric average. Therefore, the arithmetic Asian call is always *at least as expensive* as the Kemna-Vorst geometric price. OCDAP verifies this automatically as a sanity check.

4.4 Geometric Average — Kemna-Vorst (1990)

$$\sigma_g = \sigma\sqrt{\frac{2N+1}{6(N+1)}}, \qquad b_g = \frac{1}{2}\left(-\frac{\sigma^2}{6} + \sigma_g^2\right), \qquad F_g = F \cdot e^{b_g T} \tag{5}$$

5 Crack Spread Option Pricer

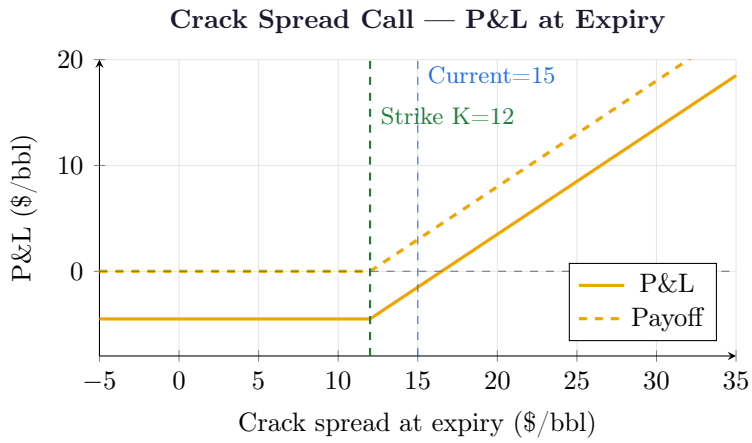
5.1 The Refining Margin



$$\text{3-2-1 Crack} = \tfrac{1}{3}\big(2 \cdot F_{\text{RBOB}} \cdot 42 + F_{\text{HO}} \cdot 42\big) - F_{\text{WTI}}$$

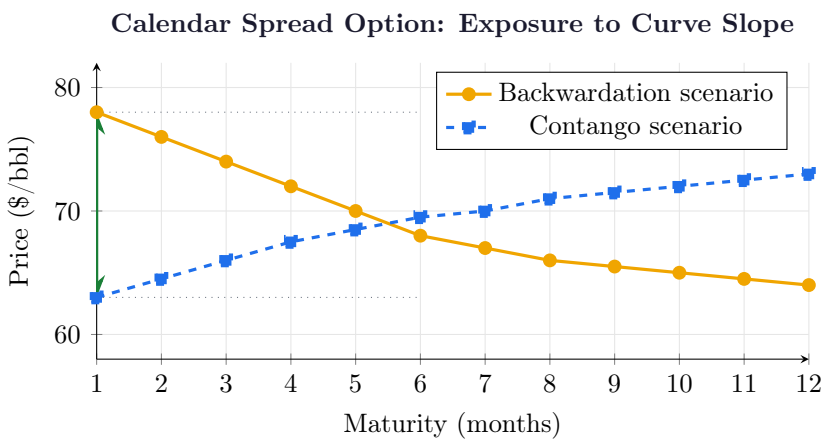
5.2 Kirk (1995) Approximation

$$\alpha = \frac{F_2}{F_2 + Ke^{-rT}}, \qquad \boxed{\sigma_{\text{Kirk}} = \sqrt{\sigma_1^2 + (\alpha\sigma_2)^2 - 2\rho\,\sigma_1\sigma_2\,\alpha}} \tag{6}$$



6 Calendar Spread Option Pricer

6.1 What a Calendar Spread Option Trades



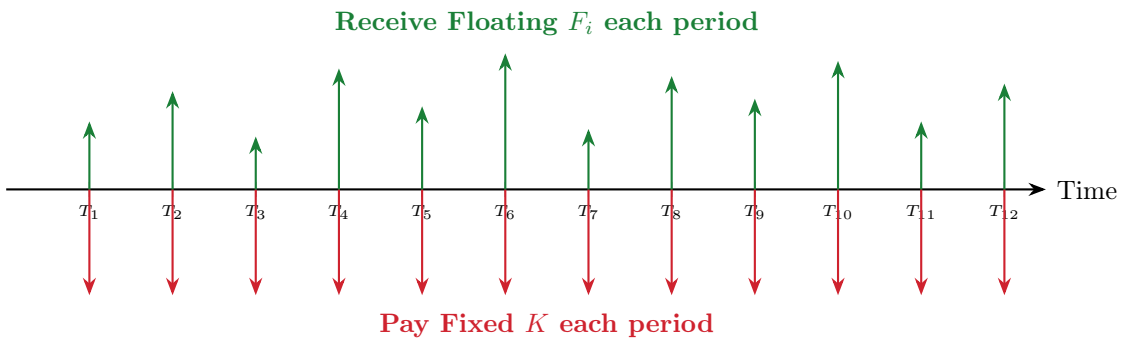
6.2 Kirk on the Term Structure

$$\alpha = \frac{F_{\text{near}}}{F_{\text{near}} + Ke^{-rT}}, \qquad \sigma_{\text{Kirk}} = \sqrt{\sigma_{\text{far}}^2 + (\alpha\,\sigma_{\text{near}})^2 - 2\rho\,\sigma_{\text{far}}\,\sigma_{\text{near}}\,\alpha} \tag{7}$$

Correlation between near and far contracts is typically $\rho \in [0.90, 0.99]$.

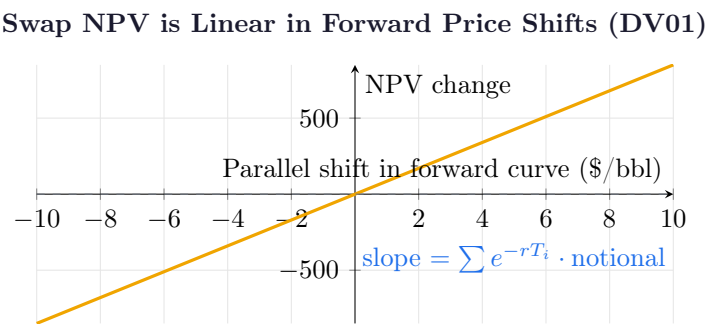
7 Commodity Swap Pricer

7.1 Cash Flow Structure



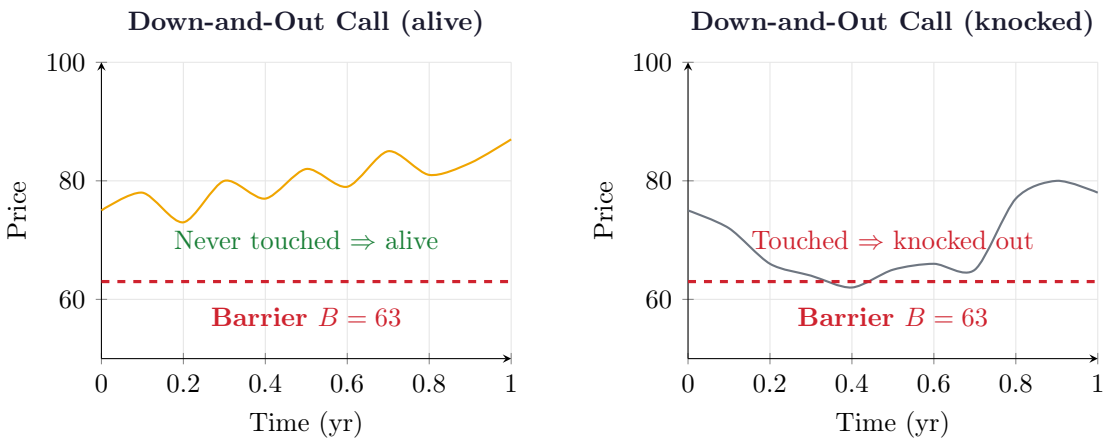
$$\boxed{\text{NPV} = \sum_{i=1}^{N} e^{-rT_i}(F_i - K) \cdot \text{notional}} \qquad K^* = \frac{\sum_{i=1}^{N} e^{-rT_i}\,F_i}{\sum_{i=1}^{N} e^{-rT_i}} \tag{8}$$

7.2 NPV Sensitivity to Forward Curve Shift

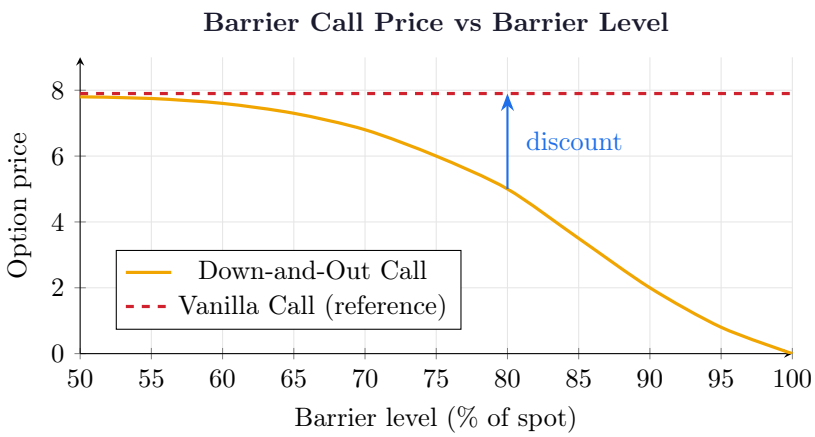


8 Barrier Option Pricer

8.1 Barrier Types

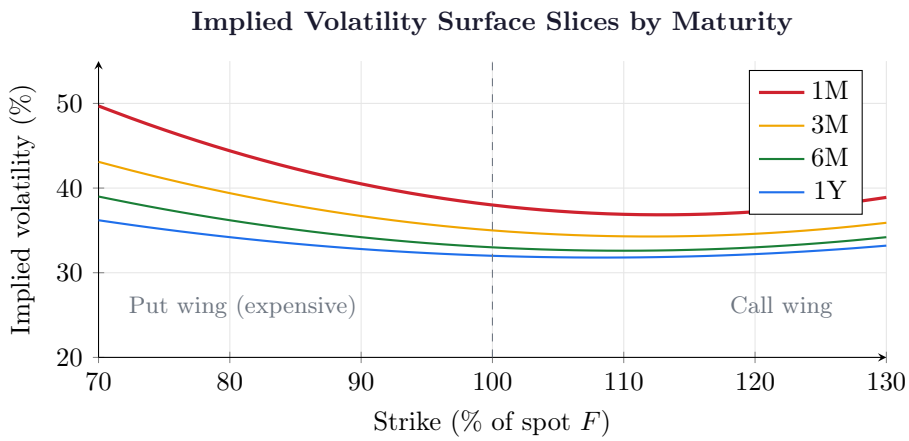


8.2 Knock Probability and Discount vs Vanilla



9 Volatility Surface

9.1 The Smile and Skew



9.2 Parametric Model

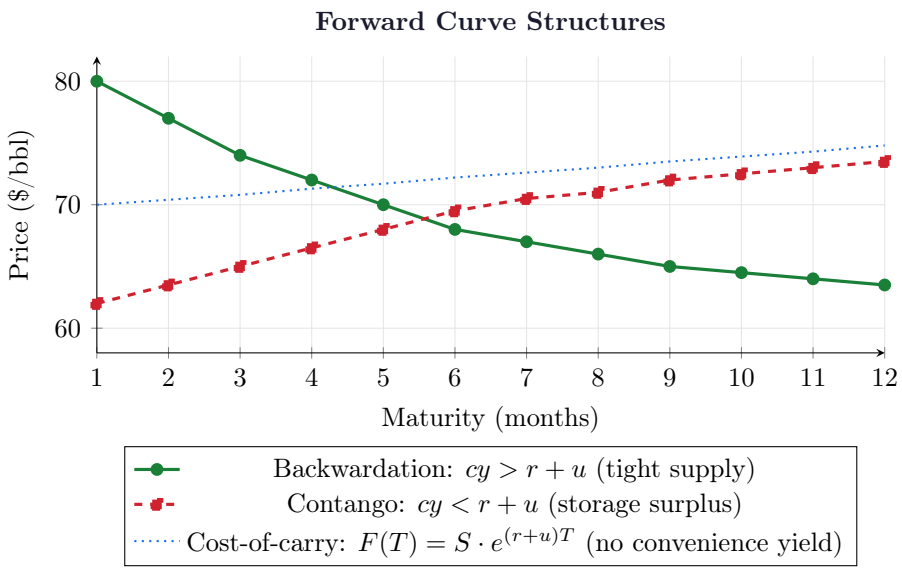
$$\sigma(K,T) = \underbrace{\sigma_{\text{ATM}}(1+\nu\sqrt{T})}_{\text{ATM term structure}} + \underbrace{\text{skew}\cdot\ln\tfrac{K}{F}}_{\text{slope}} + \underbrace{\text{curvature}\cdot\left(\ln\tfrac{K}{F}\right)^2}_{\text{wings}} \tag{9}$$

Parameter	Economic meaning	WTI typical
ATM vol σ	Base vol at the money	30%
Skew	Puts expensive relative to calls. Negative in most commodity markets.	-0.03
Curvature	How quickly vol rises in the wings. Positive = symmetric smile.	$+0.05$
Steepness ν	Term structure slope. Short-dated vol > long-dated vol.	$+0.08$

10 Forward Curve Data Layer

10.1 Cost-of-Carry Model

$$F(T) = S\cdot e^{(r+u-cy)\,T} \tag{10}$$



11 Potential Applications and Extensions

User	How OCDAP helps
Airline treasurer	Price Asian calls on jet fuel based on monthly average procurement cost. Compare vanilla vs barrier option premium to reduce hedging cost.
Crude oil refiner	Price 3-2-1 crack spread options to protect gross refining margin. Use commodity swaps to lock in a fixed crude purchase price for 12 months.
Commodity trader	Price calendar spread options to speculate on steepening or flattening. Use the vol surface to assess whether implied vol is rich or cheap relative to ATM.
Risk manager	Compute DV01 and cashflow sensitivity for swap portfolios. Quantify knock-out probability and discount on barrier options.
Student / practitioner	Apply six pricing engines to 65+ commodities with live data. All formulas, models and assumptions are transparent and documented.

11.1 Possible Quantitative Extensions

Implied volatility surface from real option chains. Yahoo Finance exposes option chain data for liquid futures (WTI, Gold, Corn) via `yfinance`. OCDAP could extract real market-implied vols per strike and maturity, and overlay them against the parametric model to reveal the smile or skew actually priced by the market.

Stochastic volatility (SABR model). The SABR model is the industry standard for capturing the volatility smile in interest rate and commodity derivatives. Replacing the flat-vol Black-76 pricer with SABR would produce more realistic prices for OTM options.

Swing options. Swing options are standard in European gas and power markets. They give the holder the right to take or deliver gas at a variable daily volume within contractual bounds. Pricing requires dynamic programming and is directly applicable to Engie, TotalEnergies, and EDF trading desks.

Monte Carlo Schwartz-Smith simulation. Using the SS calibrated parameters $(\kappa, \xi_0, \sigma_\chi, \sigma_\xi)$ as diffusion inputs, OCDAP could simulate full paths of the futures curve and price path-dependent exotic options (barrier, Asian, lookback) in a single consistent framework.

12 Conclusion

OCDAP demonstrates how a professional-grade commodity derivatives pricer can be built entirely in Python with freely available market data.

Six pricing engines cover the full spectrum of products used by commodity desks: vanilla options (Black-76), average-price options (Monte Carlo), spread options (Kirk 1995), curve slope options, fixed-for-floating swaps, and path-dependent barriers. The expiry-aware ticker logic ensures prices are always sourced from live, non-expired futures contracts.

Together with CFCAP, OCDAP provides a complete analytical stack for the commodity markets: curve analysis and signals on one side, derivatives pricing and risk on the other.